Khang Truong Giang

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My goal is to build autonomous systems that reconstruct and understand the 3D world from 2D images.

Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Ph.D. in Computer Science 09/2021 – 08/2024
Advisors: Prof. Sungho Jo, Prof. Soohwan Song. Research Interests: **3D Reconstruction**

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
M.S. in Computer Science 09/2019 – 08/2021

Hanoi University of Science and Technology (HUST) *Hanoi, Vietnam*
B.Sc. in Computer Science (School of ICT Talent Program) 09/2013 – 08/2018
GPA: 3.62 / 4.0 | Graduation with Honors (**Excellent Degree**).

Professional Experience

Visionary *South Korea*
Senior Research Engineer 08/2026 – Present

- **E2E Model Evaluation:** Built an evaluation engine for perception and closed-loop driving models to identify model weaknesses and guide further improvements.
- **Synthetic Data Generation:** Built a simulation environment using Alpasim and NuRec, with pipelines to generate synthetic data for training critical long-tail driving scenarios.

42dot (Global Software Center of Hyundai Motor Group) *South Korea*
AI Model Engineer II 07/2024 – 08/2026

- **Multi-Camera Perception:** Contributed to multi-task learning architecture development; designed and developed models for long-range object detection up to **150 m**.
- **Technical Leadership:** Led projects on 3D traffic light detection (up to **90 m**) and 2D traffic sign detection, from annotation design through model development, training, and release.
- **Data-Centric Development:** Designed manual annotation and pseudo-labeling workflows for 3D traffic lights, and the annotation process for 2D traffic signs.
- **E2E Planning:** Developed a diffusion policy for trajectory prediction, extending perception expertise to learned action generation for autonomous systems.

Neuro-Machine Augmented Intelligence Lab, KAIST *South Korea*
Graduate Researcher 09/2019 – 06/2024

- **3D Vision Research:** Conducted research in core 3D reconstruction areas, including high-precision feature matching, visual localization, and multi-view depth estimation.
- **Open-Source Research:** Authored and maintained research repositories on GitHub:
  - **CDS-MVSNet:** An accurate, efficient learning-based multi-view stereo network.
  - **TopicFM / TopicFM+:** A highly robust, efficient, and interpretable image matching model.
  - **DeViLoc:** A semi-dense visual localization system designed for robust spatial awareness.
- **Publications:** Research contributions include work at CVPR, AAAI, ICLR, and IEEE TIP.

Viettel High Technology Industries Corporation *Hanoi, Vietnam*
Software Engineer 09/2018 – 06/2019

- **Predictive Analytics:** Built a robust machine learning pipeline to analyze and predict user activity levels across the telecommunications network.
- **API Deployment:** Containerized and deployed the models as production Web APIs using Flask.

Selected Publications

B. Lee, J. Park, **Khang Truong***, and S. Song*. "Online 3D Gaussian Splatting Modeling with Novel View Selection." *Proc. 34th International Joint Conference on Artificial Intelligence (**IJCAI**)*, 2025. (**Top-tier in AI**; *Corresponding Authors)

B. Lee, J. Park, **Khang Truong**, S. Jo, and S. Song. "MVS-GS: High-Quality 3D Gaussian Splatting Mapping via Online Multi-View Stereo." *IEEE Access*, 2025. (Impact Factor: 3.6)

Khang Truong, S. Song, and S. Jo. "Learning to Produce Semi-dense Correspondences for Visual Localization." *IEEE/CVF Conf. on Computer Vision and Pattern Recognition (**CVPR**)*, 2024. (**Oral, top 3.3% — 90 of 2,719**)

Khang Truong, S. Song, and S. Jo. "TopicFM+: Boosting Accuracy and Efficiency of Topic-Assisted Feature Matching." *IEEE Transactions on Image Processing (**TIP**)*, 2024. (**Impact Factor: 15.3**)

Khang Truong, S. Song, and S. Jo. "TopicFM: Robust and Interpretable Topic-assisted Feature Matching." *AAAI Conf. on Artificial Intelligence (**AAAI**)*, 2023. (**Top-tier in AI**)

J. Mun, **Khang Truong**, Y. Lee, N. Oh, S. Huh, M. Kim, and S. Jo. "HybGrasp: A Hybrid Learning-to-Adapt Architecture for Efficient Robot Grasping." *IEEE Robotics and Automation Letters (RA-L)*, 2023. (Impact Factor: 5.2)

S. Song*, **Khang Truong***, D. Kim, and S. Jo. "Prior depth-based multi-view stereo network for online 3D model reconstruction." *Pattern Recognition*, 2023. (Impact Factor: 8.0; *Co-first Author)

Khang Truong, S. Song, and S. Jo. "Curvature-guided dynamic scale networks for multi-view stereo." *International Conference on Learning Representations (**ICLR**)*, 2022. (**Top-tier in ML/DL**)

Khang Truong, S. Song, D. Kim, and S. Choi. "Sequential Depth Completion With Confidence Estimation for 3D Model Reconstruction." *IEEE Robotics and Automation Letters (RA-L)*, 2020. (Impact Factor: 5.2)

Honors and Awards

- Peer Recognition, 42dot (Sponsored Jeju Workcation) 2024
- Outstanding Reviewer Award, CVPR 2024
- Silver Medal (Top 9%), Kaggle Image Matching Challenge 2023
- Hyundai Motor Chung Mong-Koo Scholarship for outstanding PhD students 2021 – 2024
- Full Graduate Scholarship, KAIST Master's Program 2019 – 2021
- Excellence Academic Scholarship, HUST 2018
- Lotte Scholarship for outstanding university students 2017
- Certificate of Excellence (Ranked 1st in academic cohort), HUST 2014
- Third Prize, Vietnam Mathematical Olympiad (VMO) for High School Students 2013

Academic Activities & Service

- **Conference Reviewer:** actively participate in top-tier conferences (CVPR, ICCV, AAAI, NeurIPS)
- **Journal Reviewer:** IEEE TIP (2024), IEEE TCSVT (2023), IEEE RA-L (2023), IEEE T-ASE (2024)
- **Mentorship:** Academic Mentor advising two undergraduate and master's students in research initiatives.

Technical Skills

Python, PyTorch, PyTorch Lightning, C/C++, Linux, Git.